

Bowling Score Detection from Video

DSAI 352 — Computer Vision Final Project, Spring 2026

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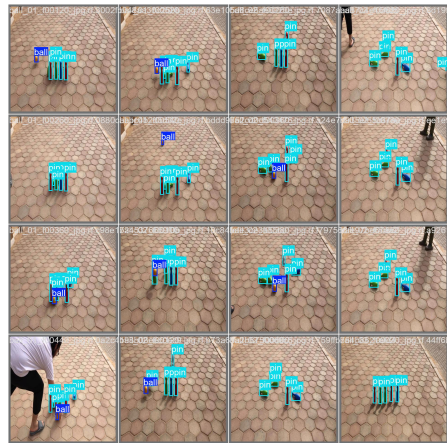
Problem Statement

Input: A mobile-recorded bowling video

- iPhone, ~30 fps, MP4 / MOV, ~30 s
- Handheld, side-on view, indoor lighting

Required output: An annotated output video showing

- Green circle marker on each pin from the moment it falls, persisting for all subsequent frames
- MM:SS timestamp label above each fallen-pin marker (the second it fell)
- Corner HUD every frame: Time: MM:SS and Pins down: N
- 2-second end-of-video summary: GAME OVER, total time, total pins down



Example annotated frame

System Pipeline

- 1 **Video Ingestion** — OpenCV reads frames from MP4/MOV at original resolution and fps.
- 2 **Detection** — Fine-tuned YOLOv8s detects ball and pin in each frame ($\text{conf} = 0.10$). Ball boxes feed the proximity gate in Stage 4.
- 3 **Tracking** — ByteTrack assigns stable integer IDs to pin detections across frames, maintaining identity through brief occlusion.
- 4 **Fall Determination** — Per-ID 3-state machine (`STANDING` $\rightarrow$ `MAYBE_FALLING` $\rightarrow$ `FALLEN`) uses aspect ratio and centroid-drop criteria, gated by ball proximity.
- 5 **Rendering** — OpenCV draws markers, timestamps, HUD, and final summary; `cv2.VideoWriter` encodes the output at original fps and resolution.

Dataset

Recording Protocol

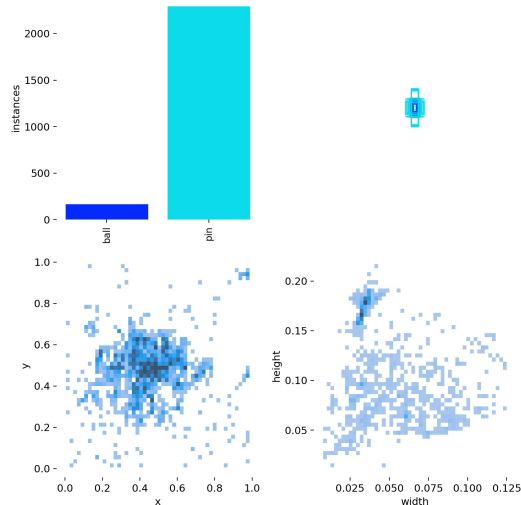
- 15 iPhone clips, ≈ 30 fps, MOV/H.264
- Same indoor environment, varied pin layouts (triangle, scattered, line)
- No controlled studio setup

Frame Extraction

- Stride 1:20 (≈ 1.5 fps effective)
- Long edge resized to 1280 px (INTER_AREA)
- **557 keyframes** extracted total

Split & Class Counts

Split	Frames	Ball / Pin
Train	407	162 / 2,291
Val	100	—
Test	50	—



Training-set label distribution

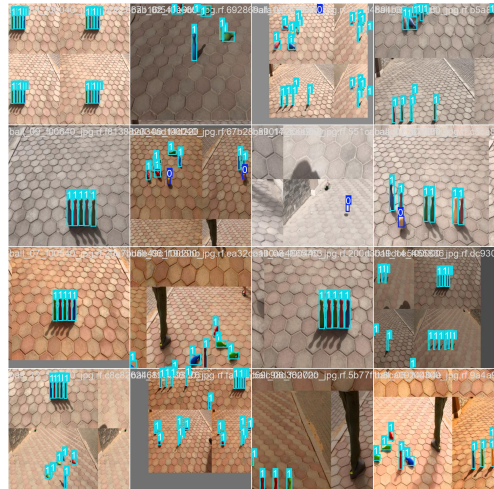
Data Remediation

Problem: Roboflow exported *polygon* labels, not axis-aligned bounding boxes. Training on the raw export caused **catastrophic convergence failure** (loss did not decrease, zero detections).

Three operations applied to 3,051 polygon annotations:

- 1 **Polygon $\rightarrow$ bbox conversion** — derive $x_{\min}$, $x_{\max}$, $y_{\min}$, $y_{\max}$ from polygon vertices; compute x_c , y_c , w , h .
- 2 **Coordinate clamping** — clamp all values to $[0, 1]$ to remove out-of-bounds vertices.
- 3 **Extreme-box filtering** — remove annotations with w or $h > 0.45$ or < 0.005 .

After remediation training converged normally.

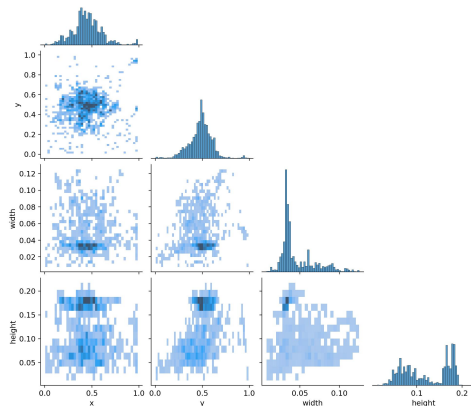


Epoch 1 training batch — correctly aligned boxes after remediation

Model Selection: Why YOLOv8s

Option	Assessment
YOLOv8n	Misses small/distant pins; insufficient recall.
YOLOv8s	Chosen. Good recall on small objects; fast training.
YOLOv8m/l/x	Negligible gain on ~ 700 images; $3\text{--}5\times$ longer training.
Faster R-CNN	Higher ceiling in principle; heavy engineering overhead not justified by dataset size.
Custom CNN	Strictly worse than a fine-tuned pre-trained model at this scale.
Classical CV	Brittle to lighting and orientation; does not satisfy the modelling requirement.

Key facts: YOLOv8s, 11M parameters, pretrained on COCO, fine-tuned on our 2-class dataset.



Label spatial distribution (training set)

Why One Class for Pins

Intuitive design: Two classes — `standing_pin` and `fallen_pin`. Class output would directly indicate fall state. **This was rejected.**

The problem with two pin classes:

- ByteTrack maintains a stable ID by matching detections across frames using IoU and appearance.
- When a pin falls, its bounding box changes shape substantially: tall and narrow → wide and short.
- If the *class label also changes* at that same frame, ByteTrack sees a disappearing `standing_pin` and a new `fallen_pin` — and assigns a **new track ID**.
- The ID switch happens at exactly the frame where the fall event must be recorded under the *original* ID.
- The state machine would never see the transition on a consistent identity.

Solution: Single `pin` class. The tracker matches through the shape change via IoU, preserving the ID. Fall detection is done geometrically on the stable ID.

Fall Detection: State Machine

Why single-frame detection fails: perspective (axial falls), motion blur (2–3 frames), occlusion during impact.

3-State machine per track ID:

STANDING $\xrightarrow{\text{criterion met}}$ MAYBE_FALLING $\xrightarrow{\text{sustained 5 consecutive frames}}$ FALLEN

If criterion fails in MAYBE_FALLING, counter resets $\rightarrow$ back to STANDING.

Fall criterion (either condition is sufficient):

$$\frac{h}{w} < 0.9 \quad \text{OR} \quad \Delta y_c > 0.5 \times h_0$$

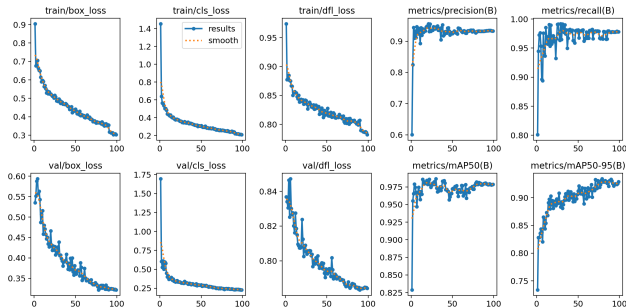
(h/w = current aspect ratio; Δy_c = centroid drop; h_0 = standing box height)

Ball-proximity gate: fall can only *begin* if the ball was within $2 \times h_{\text{pin}}$ of the pin centroid in the last 60 frames. Prevents false positives from camera shake or background motion.

Training Configuration

Parameter	Value
Model	YOLOv8s
Epochs	100
Batch	16
imgsz	640
Patience	20
lr0	0.01
cos_lr	True
Optimizer	auto
Seed	42
Hardware	Kaggle T4 GPU

Cosine LR reduces the learning rate smoothly, avoiding abrupt gradient steps that step-LR causes. Early stopping fires if val mAP does not improve for 20 epochs.



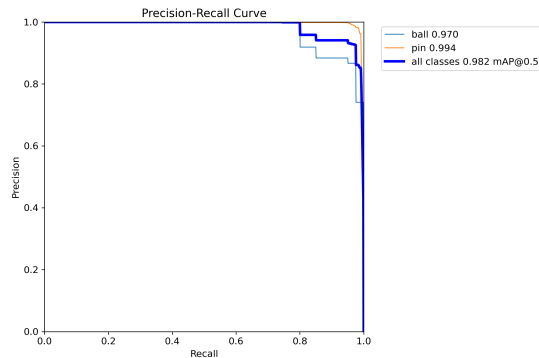
Training convergence: losses and mAP across 100 epochs

Quantitative Results

Metric	Val	Test
mAP@50	98.19%	97.19%
mAP@50-95	93.83%	91.21%
Precision	92.97%	97.69%
Recall	96.64%	88.59%
Max F1	0.9477	—
Ball AP@50	97.0%	—
Pin AP@50	99.4%	—

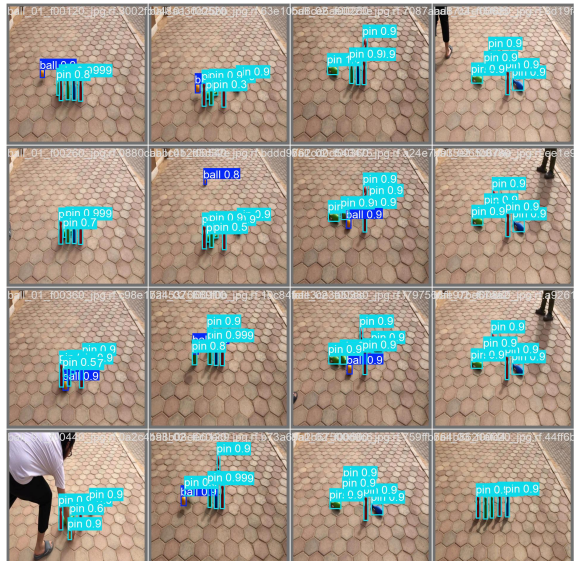
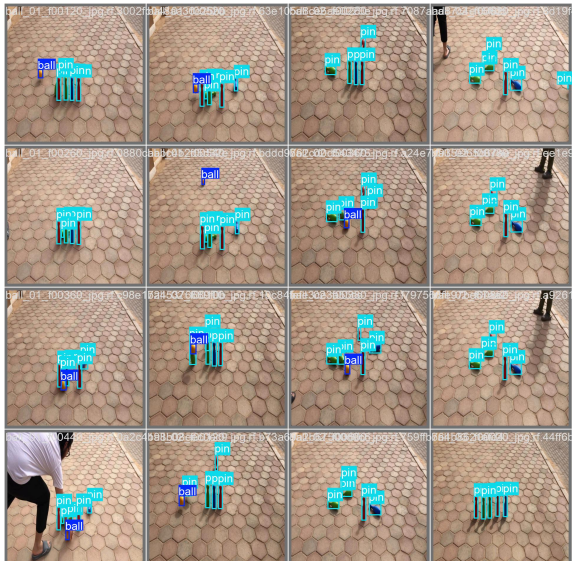
Confidence threshold = 0.10 (F1 flat from 0.10 to 0.30; chosen for maximum recall).

Test set used exactly once after all design decisions were frozen.

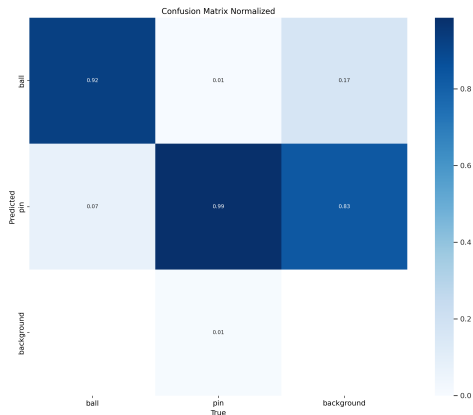


Precision-Recall curve (validation set)

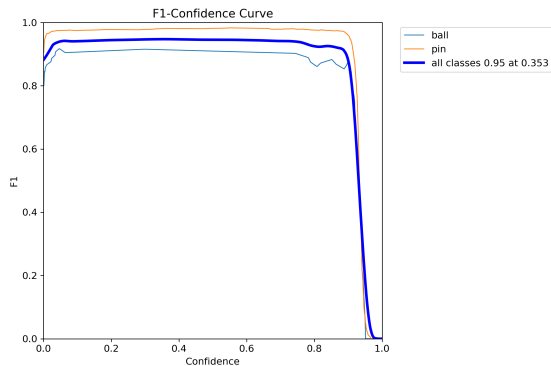
Visual Diagnostics



Confusion Matrix & F1 Curve



Normalised confusion matrix (val set)
High diagonal weight for both classes.



F1 vs. confidence threshold
Flat plateau from $\text{conf} = 0.10$ to 0.30 .

Known Limitations

- ① **Axial falls.** Pins falling toward/away from the camera change aspect ratio less. The centroid-drop signal compensates but is weaker; detection may lag by more frames.
- ② **Tracker ID switches at impact.** Heavy occlusion during ball contact can cause ByteTrack to assign a new ID. The persistent marker then attaches to the new ID rather than the original.
- ③ **Motion blur on the fall frame.** 2–3 frames of blur degrade bounding box shape. The 5-frame sustain threshold mitigates single-frame noise at the cost of a small detection lag.
- ④ **Ball-gate bypass.** If the ball detector fails for >60 consecutive frames, the proximity gate is lifted and spurious falls can be triggered.
- ⑤ **Single-environment dataset.** All 15 clips were recorded in one room. Performance on different backgrounds, lighting, or camera angles is untested.

- The fine-tuned YOLOv8s detector achieves **mAP@50 = 98.19% (val) / 97.19% (test)**, meeting and exceeding the accuracy target on both splits.
- The full pipeline — YOLOv8s + ByteTrack + 3-state fall machine + OpenCV renderer — produces the specified annotated output video with per-pin markers, MM:SS timestamps, HUD, and final summary.
- Five failure modes are documented; single-environment training data is the primary robustness bottleneck. Collecting multi-location data is the direct path to improvement.

Questions?